CMPUT 267: Machine Learning I · Fall 2026

Maximum Likelihood Estimation

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Recall: Empirical Risk Minimization

Given training data D, ERM chooses a predictor in $\mathcal{F}$ to minimize empirical risk:

$$\widehat{f}_D \in \arg\min_{f \in \mathcal{F}} \widehat{L}_D(f).$$

Expand $\widehat{L}_D(f)$ as the average loss on the individual data points:

$$\widehat{f}_D \in \arg\min_{f \in \mathcal{F}} \underbrace{\frac{1}{n} \sum_{i=1}^{n} \ell(f(\mathbf{x}_i), y_i)}_{\widehat{L}_D(f)\text{: Empirical risk}}.$$

$\ell(f(\mathbf{x}_i), y_i)$ measures how bad the prediction is for **one data point** $(\mathbf{x}_i, y_i)$.

Recall: Loss Measures Prediction Error

A **loss function** $\ell(a, y)$ measures how bad a prediction a is when the true label is y.

For regression, we have used **squared loss**:

$$\ell(a, y) = (a - y)^2.$$

- A correct prediction has loss 0.
- A larger prediction error gives a larger loss.

Recall: Bayes Predictor

When analyzing ERM and evaluating models, we used the **Bayes predictor** as the best possible reference.

It minimizes expected risk over all predictors of the available inputs:

$$f_{\text{Bayes}} \in \arg\min_f L(f), \qquad L(f) = \mathbb{E}_{\mathbf{X},Y}[\ell(f(\mathbf{X}), Y)].$$

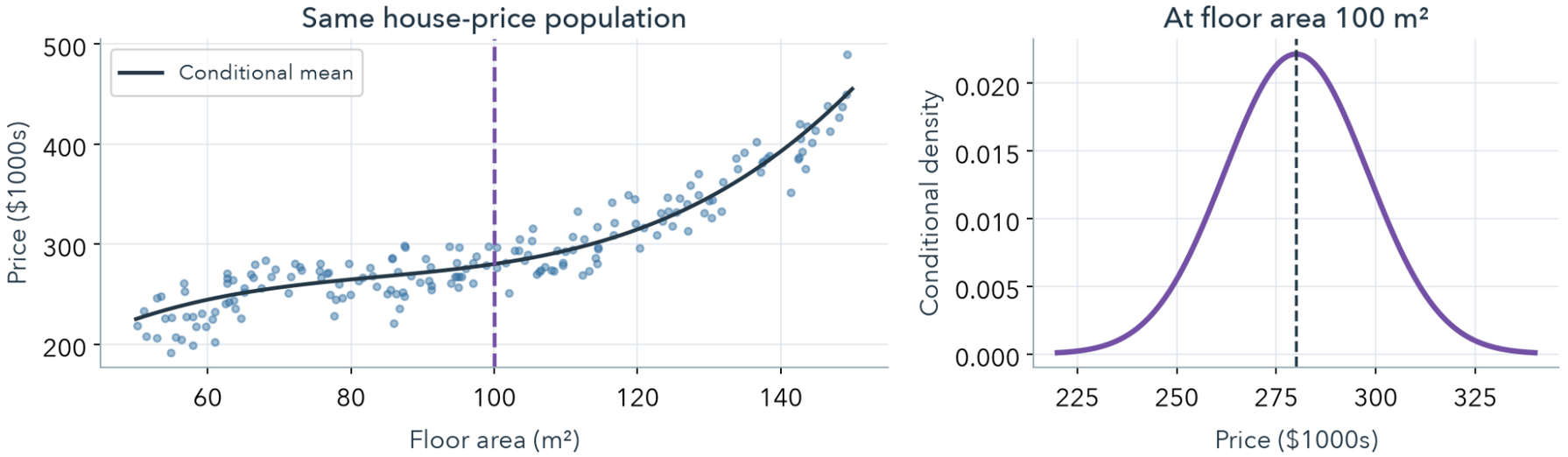
ERM minimizes empirical risk within $\mathcal{F}$. The Bayes predictor minimizes expected risk over all predictors.

? With squared loss, what is $f_{\text{Bayes}}(\mathbf{x})$? Let us derive it.

Bayes Predictor under Squared Loss

Under squared loss, the Bayes predictor is the **conditional mean**:

$$f_{\text{Bayes}}(\mathbf{x}) = \mathbb{E}[Y \mid \mathbf{X} = \mathbf{x}].$$



Deriving the Bayes Predictor

At $\mathbf{X} = \mathbf{x}$, suppose we predict a number a.

Choose a to minimize the **conditional expected squared loss**.

$$R_{\mathbf{x}}(a) = \mathbb{E}[(a - Y)^2 \mid \mathbf{X} = \mathbf{x}].$$

$$\begin{aligned} R_{\mathbf{x}}(a) &= \mathbb{E}[a^2 - 2aY + Y^2 \mid \mathbf{X} = \mathbf{x}] \\ &= a^2 - 2a\,\mathbb{E}[Y \mid \mathbf{X} = \mathbf{x}] + \underbrace{\mathbb{E}[Y^2 \mid \mathbf{X} = \mathbf{x}]}_{\text{Constant with respect to } a}. \end{aligned}$$

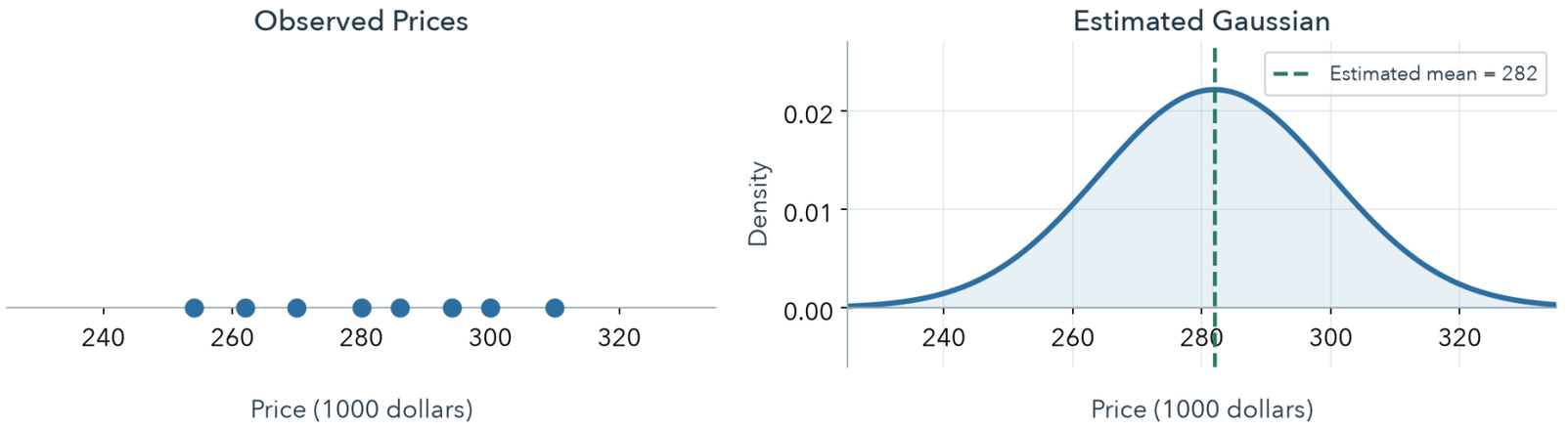
$$R'_{\mathbf{x}}(a) = 2a - 2\mathbb{E}[Y \mid \mathbf{X} = \mathbf{x}] = 0 \quad \Longrightarrow \quad f_{\text{Bayes}}(\mathbf{x}) = \mathbb{E}[Y \mid \mathbf{X} = \mathbf{x}].$$

Building a Predictor from a Distribution

For houses with features $\mathbf{X} = \mathbf{x}$, the best prediction under squared loss is the **conditional mean price**.

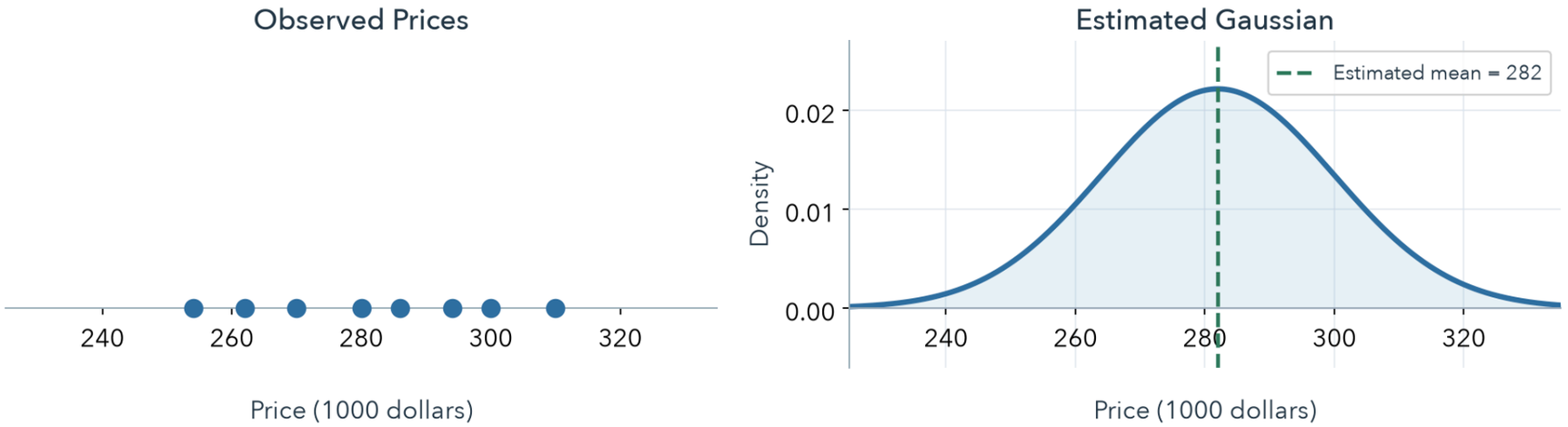
$$f_{\text{Bayes}}(\mathbf{x}) = \mathbb{E}[Y \mid \mathbf{X} = \mathbf{x}].$$

We do not know this mean, but we can **estimate it from data**: estimate the conditional distribution, then take its mean.



Today's Task: Estimate a Distribution

Observe independent samples $Z_1, \ldots, Z_n$ from the same unknown distribution.



$$\underbrace{D = (z_1, \ldots, z_n)}_{\text{Observed values}} \longrightarrow \underbrace{\widehat{p}(z)}_{\text{Estimated PMF or PDF}} .$$

This lecture introduces **maximum likelihood estimation** for this task.

Example: An Unfair Coin

We want to estimate the distribution of a coin flip. The coin may be **unfair**: heads and tails need not be equally likely.

Encode heads as 1 and tails as 0. Flip the same coin independently ten times.

$$D = (z_1, \ldots, z_{10}) = (1, 1, 1, 1, 1, 1, 1, 0, 0, 0).$$

$Z_1, \ldots, Z_{10}$ are the random flips; $z_1, \ldots, z_{10}$ are their observed values.

? What PMF should we estimate for the next flip Z?

An Intuitive PMF Estimate

We observed **seven heads and three tails** in ten flips.

Estimate each outcome's probability by how often it appeared.

$$\widehat{p}(z) = \begin{cases} 7/10, & z = 1 \quad (\text{Heads}), \\ 3/10, & z = 0 \quad (\text{Tails}). \end{cases}$$

? Can we derive this estimate mathematically, rather than just guess it?

Maximum Likelihood Estimation

Task: use the ten flips $D = (Z_1, \ldots, Z_{10})$ to estimate the coin's PMF:

$$p_Z(z \mid \theta) = \begin{cases} \theta, & z = 1 \quad (\text{Heads}), \\ 1 - \theta, & z = 0 \quad (\text{Tails}), \end{cases} \qquad 0 \leq \theta \leq 1.$$

We have already observed $\mathcal{D} = (1, 1, 1, 1, 1, 1, 1, 0, 0, 0)$.

Choose θ to give this observed dataset the **largest probability**.

$$p_D(\mathcal{D} \mid \theta) = P(Z_1 = 1, \ldots, Z_7 = 1, Z_8 = 0, Z_9 = 0, Z_{10} = 0 \mid \theta).$$

For simplicity, write this as $p_D(z_1, \ldots, z_{10} \mid \theta)$: the **likelihood** of θ for the observed data.

Likelihood and the i.i.d. Assumption

Let $D = (Z_1, \ldots, Z_n)$, where the observations are **independent and identically distributed (i.i.d.)**.

The likelihood $\mathcal{L}(\theta)$ evaluates the fixed observations given a candidate θ:

$$
\begin{aligned}
\mathcal{L}(\theta) &= p_D(z_1, \ldots, z_n \mid \theta) \\
&= p_{Z_1}(z_1 \mid \theta) \cdots p_{Z_n}(z_n \mid \theta) && \text{(Independence)} \\
&= p_Z(z_1 \mid \theta) \cdots p_Z(z_n \mid \theta) && \text{(Identical distributions)} \\
&= \prod_{i=1}^{n} p_Z(z_i \mid \theta)
\end{aligned}
$$

MLE for the Coin

Recall: for independent flips, multiply their individual probabilities.

$$\mathcal{L}(\theta) = \prod_{i=1}^{10} p_Z(z_i \mid \theta).$$

For a Bernoulli coin flip, each factor is

$$p_Z(1 \mid \theta) = P(\text{Head}) = \theta, \qquad p_Z(0 \mid \theta) = P(\text{Tail}) = 1 - \theta.$$

For a dataset D containing 7 heads and 3 tails.

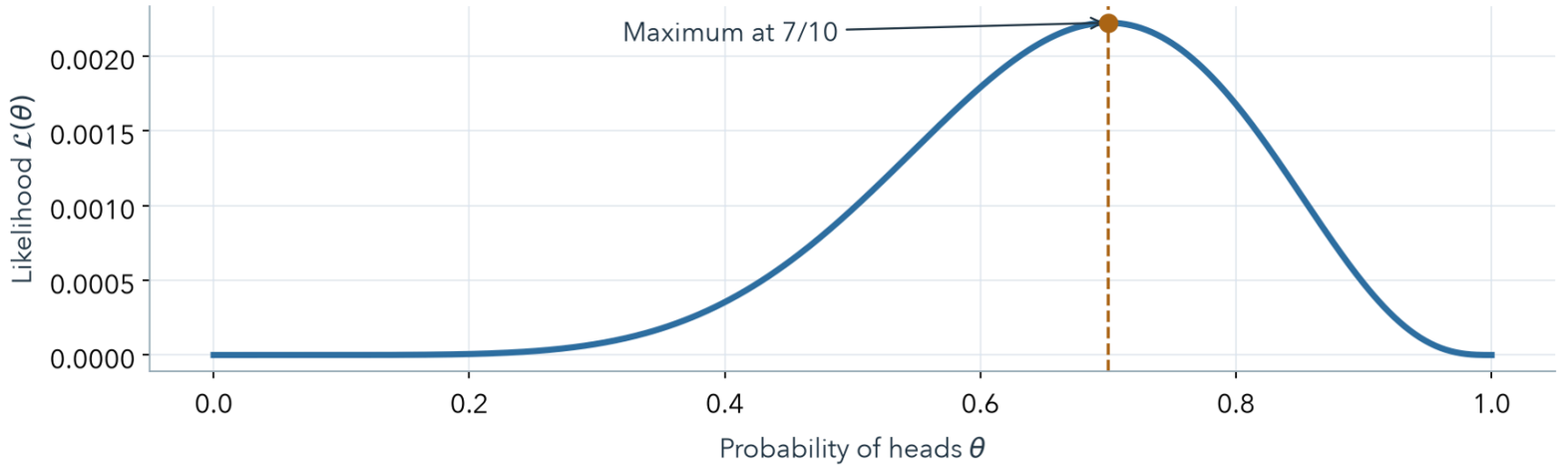
$$\mathcal{L}(\theta) = p_Z(1 \mid \theta)^7 \, p_Z(0 \mid \theta)^3 = \theta^7 (1 - \theta)^3.$$

MLE chooses $\hat{\theta}_{\text{MLE}} = \arg\max_{0 \le \theta \le 1} \theta^7 (1 - \theta)^3$.

Example: The Likelihood Curve

Recall our optimization problem for seven heads and three tails:

$$\hat{\theta}_{\text{MLE}} = \arg\max_{0 \le \theta \le 1} \theta^7 (1-\theta)^3.$$



The curve reaches its maximum at $\theta = 0.7$, so $\hat{\theta}_{\text{MLE}} = 0.7$.

The General Bernoulli Likelihood

For each coin flip, 1 means head and 0 means tail:

$$P(Z = 1) = \theta, \qquad P(Z = 0) = 1 - \theta, \qquad 0 \leq \theta \leq 1.$$

Our dataset $\mathcal{D} = (z_1, \ldots, z_n)$ contains n **independent flips**: k **ones** (heads) and $n - k$ **zeros** (tails).

$$\begin{aligned} \mathcal{L}(\theta) &= \prod_{i=1}^{n} p_Z(z_i \mid \theta) \\ &= \underbrace{\theta \cdots \theta}_{k \text{ ones}} \underbrace{(1-\theta) \cdots (1-\theta)}_{n-k \text{ zeros}} = \theta^k (1-\theta)^{n-k}. \end{aligned}$$

We want $\hat{\theta}_{\text{MLE}} = \arg\max_{0 \leq \theta \leq 1} \theta^k (1-\theta)^{n-k}$.

Trick: Minimize Negative Log Likelihood

Log preserves the maximizer; negation turns maximization into minimization.

General Form

$$\begin{aligned}
&\arg\max_\theta \prod_{i=1}^n p_Z(z_i \mid \theta) \\
&= \arg\max_\theta \sum_{i=1}^n \log p_Z(z_i \mid \theta) \\
&= \arg\min_\theta \left[-\sum_{i=1}^n \log p_Z(z_i \mid \theta) \right].
\end{aligned}$$

Bernoulli: k Heads in n Flips

$$\begin{aligned}
&\arg\max_\theta \ \theta^k (1-\theta)^{n-k} \\
&= \arg\max_\theta \ \left[k \log \theta + (n-k) \log(1-\theta) \right] \\
&= \arg\min_\theta \ \left[-k \log \theta - (n-k) \log(1-\theta) \right].
\end{aligned}$$

We often minimize **negative log likelihood (NLL)**. Products become sums, making derivatives easier.

Deriving the General Bernoulli Estimate

For k heads in n flips, recall $\mathcal{L}(\theta) = \theta^k (1-\theta)^{n-k}$.

$$\hat{\theta}_{\text{MLE}} = \arg\min_{0 \le \theta \le 1} \underbrace{\big[-k \log \theta - (n-k) \log(1-\theta) \big]}_{g(\theta)\text{: Negative log likelihood}}.$$

For $0 < k < n$, this is convex on $0 < \theta < 1$. Set the derivative to zero:

$$g'(\theta) = -\frac{k}{\theta} + \frac{n-k}{1-\theta} = 0 \quad \Longrightarrow \quad (n-k)\theta = k(1-\theta).$$

$$\boxed{\hat{\theta}_{\text{MLE}} = \frac{k}{n}} \qquad \text{For our data: } \frac{7}{10} = 0.7.$$

Check: All Flips Are Heads

Suppose all ten observed flips are heads.

? What value of θ maximizes $\mathcal{L}(\theta) = \theta^{10}$?

$\hat{\theta}_{\text{MLE}} = 1$: the largest value in $[0, 1]$.

This is an estimate from ten observations. It does **not** establish that tails are impossible.

MLE for Gaussian Mean (Example 8.2)

We observe $\mathcal{D} = (z_1, \ldots, z_n)$ from a gaussian distribution and want to estimate the population mean μ^*.

$$Z_i \overset{\text{iid}}{\sim} \mathcal{N}(\mu^*, 1), \qquad p_Z(z \mid \mu) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(z-\mu)^2}{2}\right).$$

- The mean μ^* is unknown. The variance is known and fixed at 1.
- Each candidate μ gives a different Gaussian density for the same data.

Task: choose the mean that maximizes the product of the observed density values.

MLE for Gaussian Mean: Negative Log Likelihood

Start with the maximum likelihood estimate:

$$\widehat{\mu}_{\mathrm{MLE}} = \arg\max_{\mu \in \mathbb{R}} \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(z_i - \mu)^2}{2} \right).$$

Take the negative log: turn the product into a sum, then simplify.

$$\begin{aligned} -\log \mathcal{L}(\mu) &= -\sum_{i=1}^{n} \log\left[\frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(z_i - \mu)^2}{2} \right) \right] \\ &= -\sum_{i=1}^{n} \left[-\frac{1}{2}\log(2\pi) - \frac{(z_i - \mu)^2}{2} \right] \\ &= \frac{n}{2}\log(2\pi) + \frac{1}{2}\sum_{i=1}^{n} (z_i - \mu)^2. \end{aligned}$$

The first term is constant in μ, so minimize the sum of squared deviations.

MLE for Gaussian Mean: Solution

Recall the objective after dropping the constant:

$$\widehat{\mu}_{\mathrm{MLE}} = \arg\min_{\mu \in \mathbb{R}} \tfrac{1}{2} \sum_{i=1}^{n} (z_i - \mu)^2.$$

This objective is strictly convex. Set its derivative to zero:

$$\begin{aligned} \tfrac{d}{d\mu} \left[\tfrac{1}{2} \sum_{i=1}^{n} (z_i - \mu)^2 \right] &= \sum_{i=1}^{n} (\mu - z_i) \\ &= n\mu - \sum_{i=1}^{n} z_i = 0. \end{aligned}$$

$$\boxed{\widehat{\mu}_{\mathrm{MLE}} = \tfrac{1}{n} \sum_{i=1}^{n} z_i = \bar{z}}$$

The fitted distribution is $\mathcal{N}(\bar{z}, 1)$.

MLE for Gaussian Sigma (Exercise 8.2)

Suppose that now the data $Z_1, \ldots, Z_n$ are i.i.d. with distribution

$$Z_i \sim \mathcal{N}\big(0, (\sigma^*)^2\big).$$

The mean is known and fixed at 0. The standard deviation $\sigma^* > 0$ is unknown.

? Given observations $\mathcal{D} = (z_1, \ldots, z_n)$, find the MLE of σ^*.

Start with the density:

$$p_Z(z \mid \sigma) = \frac{1}{\sqrt{2\pi}\,\sigma} \exp\left(-\frac{z^2}{2\sigma^2}\right), \qquad \sigma > 0.$$

MLE for Gaussian Sigma: Negative Log Likelihood

Start with the maximum likelihood estimate:

$$\widehat{\sigma}_{\mathrm{MLE}} = \arg\max_{\sigma>0} \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi}\,\sigma} \exp\left(-\frac{z_i^2}{2\sigma^2}\right).$$

Take the negative log: turn the product into a sum, then simplify.

$$\begin{aligned} -\log \mathcal{L}(\sigma) &= -\sum_{i=1}^{n} \log\left[\frac{1}{\sqrt{2\pi}\,\sigma} \exp\left(-\frac{z_i^2}{2\sigma^2}\right)\right] \\ &= -\sum_{i=1}^{n} \left[-\tfrac{1}{2}\log(2\pi) - \log\sigma - \frac{z_i^2}{2\sigma^2}\right] \\ &= \tfrac{n}{2}\log(2\pi) + n\log\sigma + \frac{1}{2\sigma^2}\sum_{i=1}^{n} z_i^2. \end{aligned}$$

Keep $n \log \sigma$: it depends on the parameter we are estimating.

MLE for Gaussian Sigma: Solution

Recall the objective after dropping the constant:

$$\widehat{\sigma}_{\mathrm{MLE}} = \arg\min_{\sigma>0} \left[n \log \sigma + \tfrac{1}{2\sigma^2} \sum_{i=1}^{n} z_i^2 \right].$$

Let $S = \sum_{i=1}^{n} z_i^2 > 0$. Differentiate and set the derivative to zero:

$$\begin{aligned} \tfrac{d}{d\sigma} \left[n \log \sigma + \tfrac{S}{2\sigma^2} \right] &= \tfrac{n}{\sigma} - \tfrac{S}{\sigma^3} \\ &= \tfrac{n\sigma^2 - S}{\sigma^3} = 0 \quad \Longrightarrow \quad \sigma^2 = \tfrac{S}{n}. \end{aligned}$$

$$\boxed{\widehat{\sigma}_{\mathrm{MLE}} = \sqrt{\tfrac{1}{n} \sum_{i=1}^{n} z_i^2}}$$

The derivative changes from negative to positive here, so this is the minimum.

Estimating a Distribution's Parameters

	Example 8.2	**Exercise**
Distribution	$\mathcal{N}(\mu^*, 1)$	$\mathcal{N}(0, (\sigma^*)^2)$
Known	Variance 1	Mean 0
Estimate	Mean μ^*	Standard deviation σ^*
MLE	$\widehat{\mu} = \frac{1}{n}\sum_i z_i$	$\widehat{\sigma} = \sqrt{\frac{1}{n}\sum_i z_i^2}$

In both cases, we **use data to estimate a parameter of a distribution**.

$$\mathcal{D} = (z_1, \ldots, z_n), \qquad \hat{\theta}_{\text{MLE}} = \arg\max_{\theta} \prod_{i=1}^{n} p_Z(z_i \mid \theta).$$

MLE Learner

MLE Learner

Given a observed dataset $\mathcal{D}$ of n i.i.d. feature laber pairs.

$$\mathcal{D} = \big((\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_n, y_n)\big).$$

Goal: use the dataset to estimate $p_{Y|\mathbf{X}}(y \mid \mathbf{x})$.

For each fixed $\mathbf{X} = \mathbf{x}$, estimate how the possible labels Y are distributed.

$$\widehat{p}_{Y|\mathbf{X}}(y \mid \mathbf{x}).$$

A Parameterized Conditional Distribution

To use MLE, choose a family of conditional distributions indexed by $\mathbf{w}$.

Assume a **Gaussian label distribution at each input**, with a linear mean:

$$Y_i \mid \mathbf{X}_i = \mathbf{x}_i \sim \mathcal{N}(\mathbf{x}_i^\top \mathbf{w}^*, \sigma^2).$$

- $\mathbf{w}^*$ is unknown and shared across all inputs.
- $\sigma^2 > 0$ is fixed and the same for every input $\mathbf{x}$.



This conditional Gaussian assumption belongs to our **MLE model**. Squared-loss **ERM does not require it**.

The Conditional Likelihood

For a candidate $\mathbf{w}$, the density of one observed label is

$$p_{Y|\mathbf{X}}(y_i \mid \mathbf{x}_i, \mathbf{w}) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(y_i - \mathbf{x}_i^\top \mathbf{w})^2}{2\sigma^2}\right].$$

Condition on the observed inputs. Independence gives

$$\mathcal{L}(\mathbf{w}) = \prod_{i=1}^{n} p_{Y|\mathbf{X}}(y_i \mid \mathbf{x}_i, \mathbf{w}).$$

$$\widehat{\mathbf{w}}_{\mathrm{MLE}} \in \arg\max_{\mathbf{w}} \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(y_i - \mathbf{x}_i^\top \mathbf{w})^2}{2\sigma^2}\right].$$

Minimizing the Negative Log Likelihood

Apply the same negative-log trick as in the Gaussian mean example.

$$
\begin{aligned}
-\log \mathcal{L}(\mathbf{w}) &= -\sum_{i=1}^{n} \log p_{Y|\mathbf{x}}(y_i \mid \mathbf{x}_i, \mathbf{w}) \\
&= -\sum_{i=1}^{n} \left[-\tfrac{1}{2}\log(2\pi\sigma^2) - \tfrac{(y_i - \mathbf{x}_i^\top \mathbf{w})^2}{2\sigma^2} \right] \\
&= \tfrac{n}{2}\log(2\pi\sigma^2) + \tfrac{1}{2\sigma^2}\sum_{i=1}^{n} (y_i - \mathbf{x}_i^\top \mathbf{w})^2.
\end{aligned}
$$

The first term is constant in $\mathbf{w}$. Only the squared-error sum affects the minimizer.

Solving for the MLE Parameters

Drop the constant, then multiply the objective by the positive constant $2\sigma^2/n$.

$$\widehat{\mathbf{w}}_{\mathrm{MLE}} \in \arg\min_{\mathbf{w}} \ \frac{1}{2\sigma^2} \sum_{i=1}^{n} (y_i - \mathbf{x}_i^\top \mathbf{w})^2$$

$$= \arg\min_{\mathbf{w}} \ \underbrace{\frac{1}{n} \sum_{i=1}^{n} (\mathbf{x}_i^\top \mathbf{w} - y_i)^2}_{\widehat{L}_D(\mathbf{w})\text{: Empirical squared loss}}.$$

This is the **least-squares ERM objective** we already know how to solve. Use the same least-squares solver or gradient descent.

MLE and ERM for Regression

	Gaussian MLE learner	**Squared-loss ERM learner**
Specify	$Y \mid \mathbf{X} = \mathbf{x} \sim \mathcal{N}(\mathbf{x}^\top \mathbf{w}, \sigma^2)$	$f_{\mathbf{w}}(\mathbf{x}) = \mathbf{x}^\top \mathbf{w}$, squared loss
Minimize	Negative log likelihood	Empirical squared loss

With this parameterization and conditional Gaussian assumption, **MLE gives the same learner as squared-loss ERM**.

The Gaussian assumption supplies the MLE interpretation. ERM does not need it.

Wrap-up

- **Likelihood:** evaluate the observed data under candidate parameters.
- **MLE:** choose parameters that maximize likelihood, or minimize NLL.
- **Gaussian parameters:** use data to estimate the unknown mean or standard deviation.
- **Gaussian regression:** the MLE gives least-squares parameters.

? Next lecture: how can prior knowledge about parameters produce regularization?